TITLE: EVALUATING GREEN AI INFRASTRUCTURE; AN Empirical Analysis of Subsystem Telemetry.

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ABSTRACT

The rapid growth of Artificial Intelligence(AI) has increased the demand for computing infrastructure capable of supporting increasingly intensive workloads. While AI has created opportunities for improving efficiency across different sectors, the infrastructure required to run these systems consumes significant amounts of electricity and resources. This creates a need to understand how efficiently data center infrastructure is being utilized and where improvements can be made to reduce its environmental impact. This study seeks to examine operational telemetry data to evaluate the energy efficiency and performance of AI infrastructure at the subsystem level.

The study involved data cleaning and preprocessing, exploratory data analysis, visualization and inferential statistical analysis This study uses a data-driven approach to investigate key indicators such as Computing utilization, Power Usage Effectiveness (PUE), power consumption and associated compute cost. Rather than relying only on overall datacenter energy figures, the analysis examines how individual infrastructure characteristics and workload conditions relate to energy use and operational efficiency. Exploratory data analysis, statistical comparisons, correlation analysis and anomaly detection are applied to identify patterns, relationships and unusual operating conditions within the dataset. Regional and workload level differences are also examined to understand where inefficiencies and higher energy demands are likely to occur.

The findings revealed that infrastructure efficiency is not uniform across datacenter regions. Some regions experience more intensive and variable computing activity which can increase the risk of high power demand and thermal inefficiencies. The mathematical measure for the mean CPU utilization and its variability by region is:$Ur = \frac{1}{n}\sum_{i=1}^{n} Uri$ where:

- Uri= CPU utilization observation in region
- n= number of observations
- Ur= average CPU utilization for that region.

Also in measuring how variable the CPU load is, we use the coefficient of variation which is given by:$CVr = \frac{\sigma Ur}{Ur} * 100\%$

where σUr is the standard deviation of CPU utilization in region.

INTRODUCTION

1.1 Background of the study

Artificial Intelligence (AI) has become an important part of many technologies and services that people use daily. From search engines and recommendation systems to language models and other intelligent applications, AI systems require a large amount of computing power to operate. Behind these applications are data centers containing thousands of servers that continuously process, store and transfer information. As the use of AI continues to grow, so does the amount of computing infrastructure needed to support it. This raises an important question: Are these resources being used efficiently or are data centers consuming more energy than necessary to perform their workloads. This concern has led to the growing interest in Green AI which focuses on developing and operating AI systems in a way that reduces unnecessary energy and resource consumption while still maintaining good performance. However, improving the sustainability of AI infrastructure requires more than simply looking at the total amount of electricity consumed by a data center. It is also important to understand what is happening inside the infrastructure and how computing resources are being utilized.

This is where subsystem telemetry becomes useful. Telemetry provides information about the condition and operation of computing infrastructure over time. In this study, particular attention is given to CPU utilization and data center regions. CPU utilization provides an indication of how heavily computing resources are being used. Very high utilization may reflect demanding workloads, while consistently low utilization may suggest that available resources are not being fully used. Differences between regions can provide useful insight into how workloads and computing resources are distributed.

1.2 Problem Statement

The rapid growth of AI has increased demand for high-performance data centers, leading to greater energy consumption and operational costs. Although data centers collect telemetry data such as CPU utilization, power consumption, PUE, temperature, and compute cost, these data are not always fully used to identify inefficiencies. This study analyzes subsystem telemetry to examine how computing utilization, efficiency and regional differences are related. The goal is to identify patterns such as excessive energy use and underutilization and provide insights for developing more energy-efficient, cost-effective and sustainable AI infrastructure.

1.3 Objectives of the Study

1. To examine CPU utilization patterns across different data centers to understand how computing resources are being used.
2. To compare CPU utilization between regions to identify differences in workload intensity and resource demand.
3. To identify periods of unusually high or low CPU utilization that may indicate inefficient resource use, underutilization or potential over-provisioning.
4. To use the findings from regional CPU utilization patterns to provide practical insights into improving resource efficiency and supporting more sustainable Green AI infrastructure.

1.4 Research Questions

1. How does CPU utilization vary across different datacenter regions?
2. What differences in workload intensity and resource demand can be observed between data center regions based on CPU utilization?
3. What does regional CPU utilization patterns reveal about resource efficiency and potential over-provisioning Green AI infrastructure?

1.5 Significance of the study

The significance of this study is that it uses real infrastructure telemetry to understand whether computing resources are being used efficiently. By examining CPU utilization, power consumption and regional differences, the study can identify patterns of underutilization and high resource demand that may contribute to unnecessary energy consumption. These insights can support better resource management and contribute to the development of more energy-efficient and sustainable AI data center infrastructure.

1.6 Scope of the study

This study focuses on evaluating the efficiency of AI data center infrastructure using real subsystem telemetry data. The analysis is mainly centered on CPU utilization and data center regions, while also considering related measures such as power consumption, PUE and compute cost. These variables are used to understand how computing resources area being utilized and to identify patterns that may suggest high demand, underutilization and possible energy inefficiencies.

METHODOLOGY

2.1 Research Design

This study used a quantitative, empirical research method based on secondary data analysis. These methodologies were used because this study works well with numerical telemetry data such as CPU utilization, power consumption and data center region. Statistical analysis and comparative visualizations are used to identify patterns and relationships in the data.

The empirical research is employed to understand how data center infrastructure actually behaves.

2.2 Data Source

This study uses secondary data obtained from data center subsystem telemetry. The data source for this project is the Green AI Hyperscale Data Center Telemetry Dataset sourced from Kaggle. It is a synthetic, large-scale dataset containing 100000 rows and 35 columns designed to simulate the operational behavior and infrastructure layers of modern cloud data centers hosting high performance computing and AI workloads.

The dataset includes variable such as

- Server type
- Cooling type
- Datacenter region
- CPU utilization
- Storage utilization
- Memory utilization
- Inlet temperature
- Exhaust temperature

The dataset was imported into R studio for cleaning, exploration and statistical analysis.

2.3 Programming Language Adopted

The analysis was conducted using **R Programming Language** within **RStudio**.

2.1: R Packages Used

Packages	**Purpose**
tidyverse	Data Manipulation
dplyr	Data cleaning and transformation
Psych	Descriptive Statistics
ggplot2	Data Visualization
plotly	Interactive Visualization
corrplot	Correlation plots
rstatix	Statistical tests

2.4 Research Methodology

This study adopted the **Cross-Industry Standard Process for Data Mining (CRISP-DM)** methodology.

CRISP-DM provides a structured framework for carrying out data mining and statistical analysis projects. The methodology consists of six iterative phases:

- Business Understanding
- Data Understanding
- Data Preparation
- Modeling
- Evaluation
- Deployment

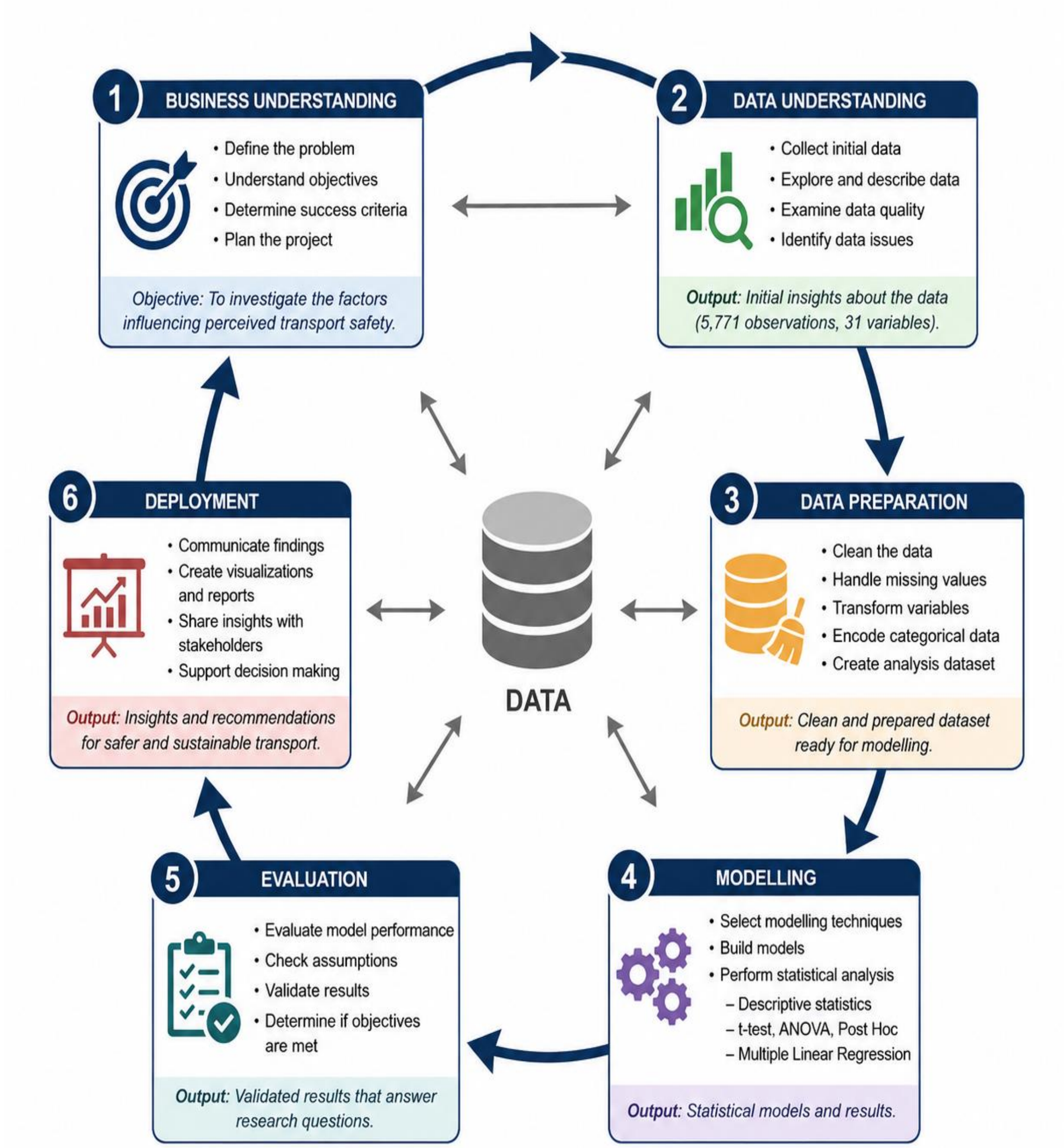


Fig 2.1 illustrates the CRISP-DM framework adopted in this study

2.4.1 Business Understanding

The business understanding of this project is to determine whether AI datacenter resources are being used efficiently and to identify where energy, computing capacity or operational costs may be unnecessarily high. The findings can help data center operators improve resource utilization, reduce energy waste and operating costs and also support more sustainable AI infrastructure.

2.4.2 Data Understanding

This phase focused on getting a clear understanding of the dataset before carrying out any cleaning or transformation. The data was examined through the following steps:

- Viewing the dataset
- Examine the dimensions
- Checked data types
- Identifying the dataset variable
- Identifying missing values
- Generating summary statistics
- Checking for duplicate observation

The dataset contains both numerical and categorical variables with numerical variables representing CPU utilization, compute cost and other telemetry measurements. These are values that can be measured and used in statistical analysis whereas Categorical variables includes datacenter regions such as APAC, EU, ME and NA

2.4.3 Data Preparation

The data preparation stage focused on making sure that the Green AI infrastructure dataset was suitable for further analysis. The dataset was first examined to understand its dimensions, variable names and data types. The data was checked for missing values and duplicate observations to identify any potential issues that could affect the results.

Summary statistics were also generated to understand the distribution and range of the numerical variables. Particular attention was given to variables such as CPU utilization, PUE as these were central to evaluating the efficiency of AI infrastructure.

I. Handling Missing Values

The colSums (is.na(green_ai_datacenter)) evaluates every cell to check if it is missing (NA) and then sums the TRUE values for each column. This showed a list of my dataset's column names with a 0 underneath each one. This implies there are zero missing values (NAs) across all columns in the green_ai_datacenter dataset as this data is clean.

```
# Checking for Missing Values
# Missing Column Values
colSums(is.na(green_ai_datacenter))
sum(is.na(green_ai_datacenter))

# Checking for duplicates
sum(duplicated(green_ai_datacenter))
```

Fig 2.2 Snippet of the code showing handling missing values and checking duplicates

2.5 EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was conducted to summarize the characteristics of the dataset and identify patterns. The following descriptive statistics were generated:

- Mean
- Medium
- Standard deviations
- Frequency distributions
- Percentages

Visualizations produced include

- Bar Chart
- Box plot
- Correlation matrix

2.6 DESCRIPTIVE STATISTICS

The numerical variables analyzed:

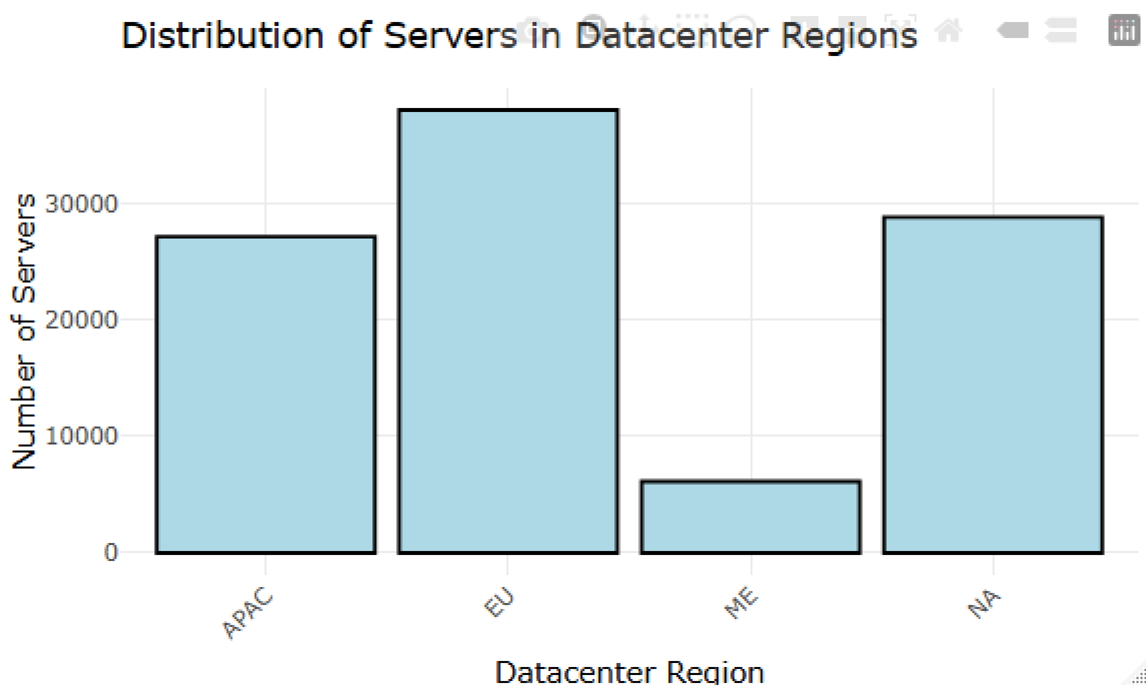
- APAC
- EU
- ME
- NA

2.3: Descriptive Statistics of Numerical Variables

Variable	Mean	Median	Standard deviation
APAC	47.3	47.4	18.8
EU	47.2	47.2	18.8
ME	47.0	46.6	18.9
NA	47.1	47.0	18.8

Across the four regions, the mean and median are nearly identical which is (47.0 to 47.4). There is also uniformity across all four regions. APAC has the highest average mean of 47.3, while ME has the lowest with 47.0.

2.6.1 Frequency Distribution



This is a vertical bar illustrating the distribution of servers across different datacenter regions.

EU holds the highest concentration of server resources, peaking at 37972. This suggest that the primary workloads are heavily centered in this region which is then followed by NA with 28801 servers then APAC with 27133 servers and the lowest being ME with 6094 servers represent a minimal baseline allocation, maintaining the lowest operational capacity. This pronounced disparity between the ME region and the broader global datacenters highlights a highly localized distribution strategy, which likely mirrors regional market demand. Compliance structures and user base distributions.

2.6.2 CPU Utilization Distribution Analysis

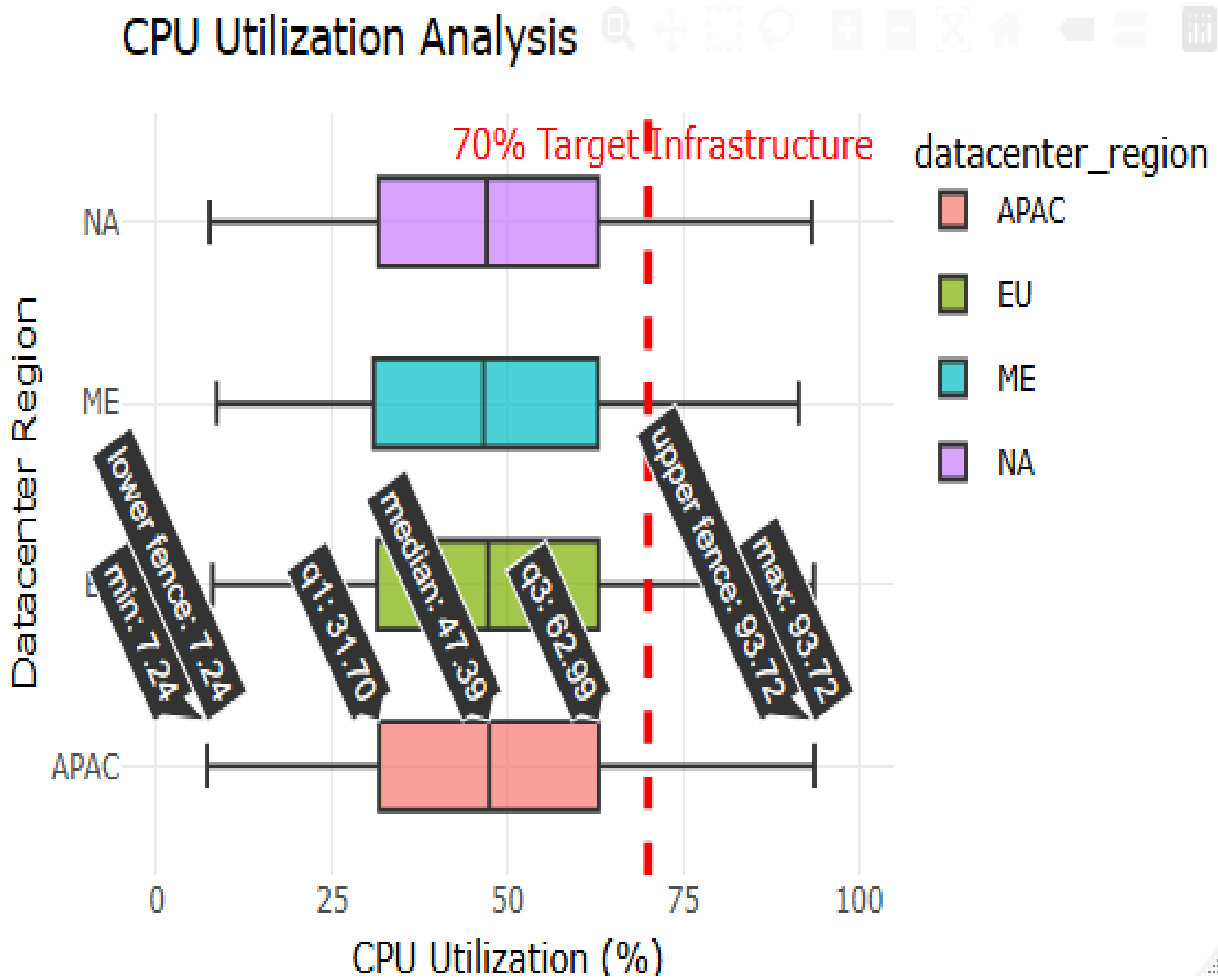
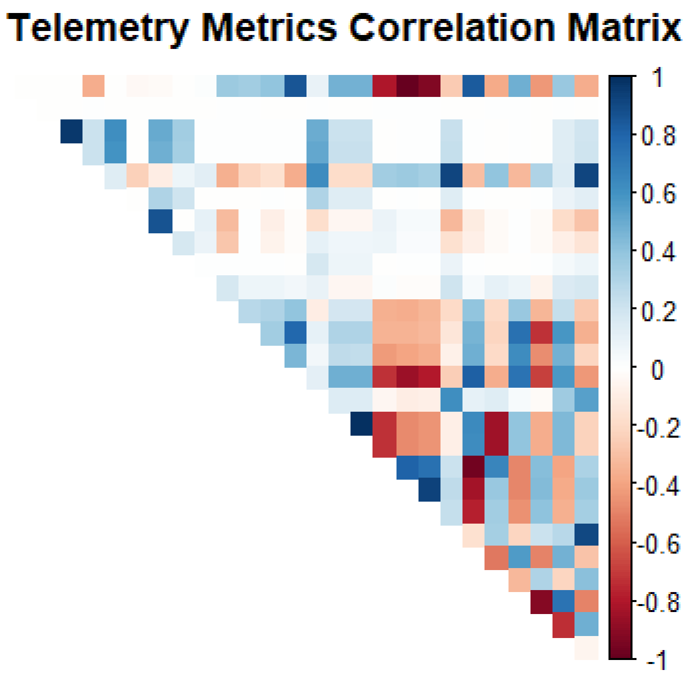


Fig 2.6.2 shows a boxplot analysis evaluating the distribution of CPU Utilization (%) across the four primary datacenter regions (NA, ME, EU, APAC) relative to 70% target infrastructure utilization threshold which is represented by a red dashed line. The visualization reveals a highly uniform operational profile across all datacenter regions. The median CPU utilization for all four datacenter regions is consistently centered between 40% and 45%, indicating stable baseline

workloads globally. Furthermore, the interquartile ranges (IQRs) display identical dispersion patterns. While bulk of the data points over 75% safely reside below the 70% optimization threshold, the upper whiskers extend up to approximately 90% across all regions. This indicates that while average resource consumption is well-managed, all datacenters experience comparable peak surge periods that breach the target threshold, highlighting a potential need for workload balancing strategies during peak hours.

2.6.3 Telemetry Metrics Correlation Matrix



To understand the operational efficiency of our infrastructure, a telemetry metrics correlation matrix mapping CPU utilization against datacenter regions was generated. By examining the intersection of computational demand and geographical location, One can empirically evaluate how effectively infrastructure balances AI workloads against grid-level sustainability goals.

The deep blue square matching with CPU utilization is the first region. This tells us that Region 1 is currently bearing the brunt of our AI model training. While this means high hardware efficiency, it also flags the first region as the primary target for energy-saving optimizations, since high CPU used directly drives up local power.

2.7 DATA ANALYSIS

Analyzing regional server counts alongside CPU utilization profiles exposes a critical structural challenge in sustainable computing: the systematic environmental cost of over-provisioning. While the hardware footprint varies drastically between territories which ranges from a massive hub of 37972 servers in the EU to a minimal baseline of 6094 servers in the Middle East. Every single region operates with a median CPU utilization floating between 40% and 45%. This flat consistency reveals that despite vastly different geographic markets and physical infrastructure scales, the underlying operational approach remains uniform and highly inefficient.

This baseline underutilization represents a severe sustainability deficit. Modern server hardware draws a disproportionate amount of base power simply staying power on. Running thousands of servers at less than half capacity means a massive shares of total power consumption goes towards maintaining idle hardware rather than performing meaningful computation. The telemetry proves that the target infrastructure threshold of 70% where a datacenter achieves optimal performance. To prevent system crashes, intense workload surges, entire regions are permanently over-provisioned. The data clearly shows that static resource allocation forces an aggressive trade-off.

2.8 DISCUSSION OF FINDINGS

The findings from the telemetry data highlights a major gap between physical infrastructure size and actual energy efficiency. Although server counts differ drastically by region, their operational behavior is identical. Every region runs at a middle of the road median CPU utilization of just 40% to 45%.

From a sustainability perspective, this constant underutilization is a significant problem. Servers draw a massive amount of base power just by being turned on, even when they are barely doing any work. Keeping thousands of machine running at less than half capacity means a huge amount of electricity is wasted on idle infrastructure, failing to hit the green target zone of 70% efficiency. The data explains why this happens, the upper spikes reach 90% utilization across all regions. This proves that datacenters are intentionally oversized and over-provisioned just to handle brief, heavy spikes in demand.

Ultimately, these findings show that static infrastructure setups are too wasteful for Green AI standards. To curtail this, datacenters need to move away from static setups and use dynamic scheduling by consolidating active workloads onto fewer servers to keep them running at a highly efficient 70% load, while putting idle servers into power-saving sleep modes.

2.9 RECOMMENDATIONS

To resolve the energy waste caused by low average utilization and high peak surges (90%), the datacenter infrastructure should transition from static management to dynamic, telemetry- driven mechanisms:

- Workload Consolidation: Automatically pack low-utilization AI jobs onto fewer physical servers to push active nodes into the efficient 70% target utilization zone.
- Dynamic Power States: Automatically transition unneeded idle servers into deep sleep modes during off-peak hours to eliminate baseline power drain.
- Predictive Auto-Scaling: Use machine learning to forecast the 90% CPU spikes before they happen, allowing the system to maintain a lean baseline without risking crashes.
- Carbon-Aware Scheduling; Shift flexible, non-urgent AI training workloads to times or regions with the highest with the highest renewable grid energy.

3.0 CONCLUSION

This empirical analysis of subsystem telemetry highlights a critical point in the pursuit of sustainable computing. The substantial environmental overhead caused by static infrastructure management. The telemetry data confirms that while the physical allocation of server assets varies significantly by region, operational efficiency remains uniformly low.

Across all global territories, median CPU utilization resides at an inefficient 40% to 45%, failing to reach the optimal 70% infrastructure utilization target. This mismatch reveals that massive amounts of electrical power are consistently wasted maintaining idle or underutilized hardware. The telemetry further demonstrates that this over-provisioning is a defensive choice. Datacenters are deliberately oversized to absorb severe, intermittent workload surges that spike toward 90% utilization. Under traditional static frameworks, ensuring system reliability during these brief peaks requires running an expanded, energy-inefficient baseline footprint. This operational trade-off is fundamentally incompatible with Green AI objectives, as the constant idle power draw directly compromises the carbon-reduction benefits of optimized AI algorithms.

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